Where am I?

Campus Building Recognition with Deep Learning

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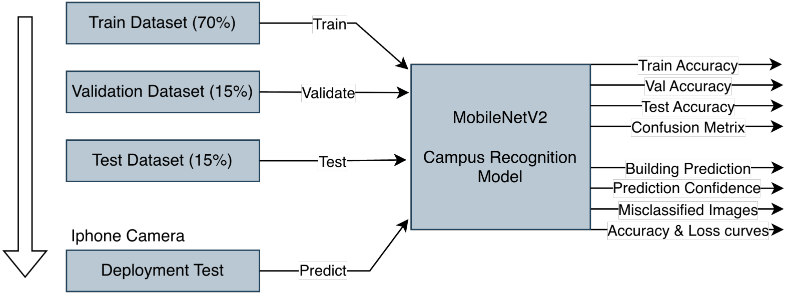
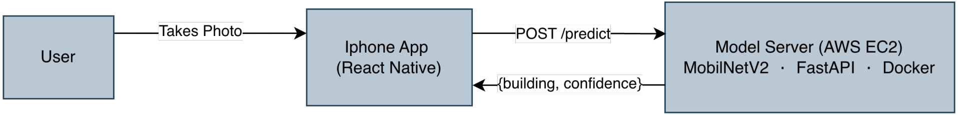
Deep Learning Capstone

- Collected a custom campus dataset and fine-tuned **MobileNetV2 to 94% test accuracy**.
- Built a complete **full-stack deployment pipeline**.
- Developed and tested a working application on a real **iPhone**.

Context Diagrams





Project Goals Achieved

- **Custom Image Classification Model**
  Collected a real-world LTU campus dataset and fine-tuned MobileNetV2 to accurately recognize 8 campus buildings.

- **Production-Ready Cloud API Deployment**
  Deployed the trained model as a Dockerized FastAPI backend on AWS EC2, enabling live prediction requests from external devices.

- **Functional iPhone Mobile Application**
  Built a React Native iOS application that performs real-time campus building recognition directly from the phone camera.

Results, Conclusion, and Future Work

Quantitative Results

- Test Accuracy: **94.02%**
- 8 LTU building classes
- MobileNetV2 outperformed baseline CNN

Advantages

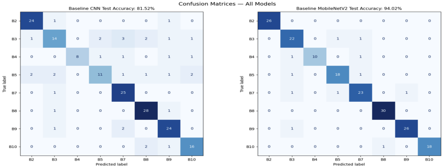
- No GPS required
- Complete end-to-end pipeline

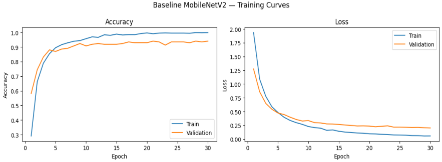
Future Work

- Release on Apple App Store

Key Takeaway: From raw campus photos to a live iPhone app — a fully deployed deep learning system on cloud.

Model Visualizations





True: Building 7
Pred: Building 9

True: Building 10
Pred: Building 8